

Geometry & Trigonometry

ANSWER KEY



Area and perimeter of basic shapes

- 1 Find the area of a triangle with base 12 and height 7.

$$\text{Area} = \frac{1}{2}(12)(7) = 42.$$

- 2 Find the area and circumference of a circle with radius 5, in terms of π .

$$\text{Area} = \pi r^2 = 25\pi. \text{ Circumference} = 2\pi r = 10\pi.$$

The Pythagorean theorem

- 3 A right triangle has legs 6 and 8. Find the length of the hypotenuse.

$$c = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$$

- 4 A ladder 13 feet long leans against a wall, with its base 5 feet from the wall. How high up the wall does the ladder reach?

$$\text{Using the Pythagorean theorem: } h = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12 \text{ feet.}$$

Volume and surface area of solids

- 5 Find the volume of a rectangular prism with dimensions 4 by 5 by 6.

$$\text{Volume} = 4 \times 5 \times 6 = 120 \text{ cubic units.}$$

- 6 Find the volume of a cylinder with radius 3 and height 10, in terms of π .

$$\text{Volume} = \pi r^2 h = \pi(9)(10) = 90\pi.$$

Similar triangles and proportional reasoning

- 7 Triangle ABC is similar to triangle DEF , with $AB = 6$, $DE = 9$, and $BC = 8$. Find EF .

$$\text{The scale factor is } \frac{DE}{AB} = \frac{9}{6} = 1.5. \text{ So } EF = 1.5 \times BC = 1.5 \times 8 = 12.$$

- 8 A tree casts a shadow 20 feet long, while a 6-foot person standing nearby casts a shadow 4 feet long. Find the height of the tree using similar triangles.

$$\text{Set up a proportion: } \frac{\text{tree height}}{20} = \frac{6}{4}. \text{ So tree height} = \frac{6}{4} \times 20 = 30 \text{ feet.}$$

Right triangle trigonometry

- 9 In a right triangle, the angle θ has an opposite side of length 5 and hypotenuse 13. Find $\sin \theta$, $\cos \theta$, and $\tan \theta$.

The adjacent side is $\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$. $\sin \theta = \frac{5}{13}$, $\cos \theta = \frac{12}{13}$, $\tan \theta = \frac{5}{12}$.

- 10 A 20-foot ladder leans against a wall, making a 60° angle with the ground. Find the height the ladder reaches on the wall, to the nearest tenth.

Height = $20\sin(60^\circ) = 20\left(\frac{\sqrt{3}}{2}\right) = 10\sqrt{3} \approx 17.3$ feet.

Angle relationships and circle theorems

- 11 Two parallel lines are cut by a transversal. If one angle measures 65° , find the measure of its corresponding angle and its co-interior (same-side interior) angle.

The corresponding angle is equal: 65° . The co-interior angle is supplementary to it (they sum to 180°): $180^\circ - 65^\circ = 115^\circ$.

- 12 A central angle in a circle measures 80° . Find the measure of the inscribed angle that subtends the same arc.

An inscribed angle is always half the central angle subtending the same arc: $\frac{80^\circ}{2} = 40^\circ$.

The unit circle and radian measure

- 13 Convert 120° to radians, in terms of π .

$$120^\circ \times \frac{\pi}{180^\circ} = \frac{2\pi}{3}.$$

- 14 Find the exact value of $\cos\left(\frac{\pi}{3}\right)$ using the unit circle.

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}.$$

Applying geometry and trigonometry to real-world contexts

- 15 A surveyor stands 100 feet from the base of a building and measures the angle of elevation to the top as 35° . Find the height of the building, to the nearest foot.

Height = $100\tan(35^\circ) \approx 100(0.7002) \approx 70$ feet.

Composite area and shaded regions

- 16** A square with side length 10 has a circle of radius 4 cut out of its center. Find the remaining area, in terms of π .

Square area: $10^2 = 100$. Circle area: $\pi(4)^2 = 16\pi$. Remaining area: $100 - 16\pi$.

- 17** Find the area of a trapezoid with parallel sides 6 and 10, and height 5.

$$\text{Area} = \frac{1}{2}(b_1 + b_2)h = \frac{1}{2}(6 + 10)(5) = \frac{1}{2}(16)(5) = 40.$$

Coordinate geometry: distance and midpoint

- 18** Find the distance between the points (1, 2) and (7, 10).

$$\text{Distance} = \sqrt{(7 - 1)^2 + (10 - 2)^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$$

- 19** Find the midpoint of the segment connecting (−3, 5) and (9, −1).

$$\text{Midpoint} = \left(\frac{-3 + 9}{2}, \frac{5 + (-1)}{2} \right) = \left(\frac{6}{2}, \frac{4}{2} \right) = (3, 2).$$

Equation of a circle

- 20** Write the equation of a circle with center (2, −3) and radius 5.

$$(x - 2)^2 + (y + 3)^2 = 25.$$

- 21** Find the center and radius of the circle $(x + 1)^2 + (y - 4)^2 = 36$.

$$\text{Center: } (-1, 4). \text{ Radius: } \sqrt{36} = 6.$$

Arc length and sector area

- 22** A circle has radius 6. Find the arc length of a 60° sector, in terms of π .

$$\text{Arc length} = \frac{\theta}{360^\circ} \times 2\pi r = \frac{60}{360} \times 2\pi(6) = \frac{1}{6} \times 12\pi = 2\pi.$$

- 23** Find the area of a 90° sector of a circle with radius 8, in terms of π .

$$\text{Sector area} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{90}{360} \times \pi(64) = \frac{1}{4} \times 64\pi = 16\pi.$$

Multiple choice — easy

24 A right triangle has legs 20 and 48. Find the length of the hypotenuse.

- a. 53
- b. 68
- c. 52 ✓
- d. 51

$$c = \sqrt{20^2 + 48^2} = \sqrt{400 + 2304} = \sqrt{2704} = 52.$$

25 A right triangle has legs 12 and 16. Find the length of the hypotenuse.

- a. 28
- b. 19
- c. 21
- d. 20 ✓

$$c = \sqrt{12^2 + 16^2} = \sqrt{144 + 256} = \sqrt{400} = 20.$$

26 A right triangle has legs 18 and 80. Find the length of the hypotenuse.

- a. 81
- b. 83
- c. 98
- d. 82 ✓

$$c = \sqrt{18^2 + 80^2} = \sqrt{324 + 6400} = \sqrt{6724} = 82.$$

27 A right triangle has legs 15 and 36. Find the length of the hypotenuse.

- a. 40
- b. 51
- c. 38
- d. 39 ✓

$$c = \sqrt{15^2 + 36^2} = \sqrt{225 + 1296} = \sqrt{1521} = 39.$$

28 A right triangle has legs 10 and 24. Find the length of the hypotenuse.

a. 26 ✓

b. 25

c. 27

d. 34

$$c = \sqrt{10^2 + 24^2} = \sqrt{100 + 576} = \sqrt{676} = 26.$$

29 A right triangle has legs 9 and 12. Find the length of the hypotenuse.

a. 14

b. 21

c. 15 ✓

d. 16

$$c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$$

30 A right triangle has legs 24 and 32. Find the length of the hypotenuse.

a. 40 ✓

b. 39

c. 41

d. 56

$$c = \sqrt{24^2 + 32^2} = \sqrt{576 + 1024} = \sqrt{1600} = 40.$$

31 A right triangle has legs 27 and 120. Find the length of the hypotenuse.

a. 123 ✓

b. 124

c. 122

d. 147

$$c = \sqrt{27^2 + 120^2} = \sqrt{729 + 14400} = \sqrt{15129} = 123.$$

32 A right triangle has legs 36 and 160. Find the length of the hypotenuse.

a. 164 ✓

b. 165

c. 163

d. 196

$$c = \sqrt{36^2 + 160^2} = \sqrt{1296 + 25600} = \sqrt{26896} = 164.$$

33 A right triangle has legs 21 and 72. Find the length of the hypotenuse.

a. 76

b. 74

c. 75 ✓

d. 93

$$c = \sqrt{21^2 + 72^2} = \sqrt{441 + 5184} = \sqrt{5625} = 75.$$

34 A right triangle has legs 24 and 45. Find the length of the hypotenuse.

a. 50

b. 52

c. 51 ✓

d. 69

$$c = \sqrt{24^2 + 45^2} = \sqrt{576 + 2025} = \sqrt{2601} = 51.$$

35 A right triangle has legs 40 and 42. Find the length of the hypotenuse.

- a. 57
- b. 59
- c. 82
- d. 58 ✓

$$c = \sqrt{40^2 + 42^2} = \sqrt{1600 + 1764} = \sqrt{3364} = 58.$$

36 A right triangle has legs 8 and 15. Find the length of the hypotenuse.

- a. 23
- b. 17 ✓
- c. 18
- d. 16

$$c = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17.$$

37 A right triangle has legs 6 and 8. Find the length of the hypotenuse.

- a. 14
- b. 11
- c. 9
- d. 10 ✓

$$c = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$$

38 A right triangle has legs 14 and 48. Find the length of the hypotenuse.

- a. 51
- b. 49
- c. 50 ✓
- d. 62

$$c = \sqrt{14^2 + 48^2} = \sqrt{196 + 2304} = \sqrt{2500} = 50.$$

Multiple choice — medium

39 Find the distance between the points $(-5, -1)$ and $(-35, -73)$.

- a. 79
- b. 102
- c. 77
- d. 78 ✓

$$\text{Distance} = \sqrt{(-35 - (-5))^2 + (-73 - (-1))^2} = \sqrt{(-30)^2 + (-72)^2} = \sqrt{6084} = 78.$$

40 Find the distance between the points $(-3, -8)$ and $(2, -20)$.

- a. 12
- b. 17
- c. 13 ✓
- d. 14

$$\text{Distance} = \sqrt{(2 - (-3))^2 + (-20 - (-8))^2} = \sqrt{(5)^2 + (-12)^2} = \sqrt{169} = 13.$$

41 Find the distance between the points $(3, -3)$ and $(-12, -39)$.

- a. 51
- b. 38
- c. 40
- d. 39 ✓

$$\text{Distance} = \sqrt{(-12 - (3))^2 + (-39 - (-3))^2} = \sqrt{(-15)^2 + (-36)^2} = \sqrt{1521} = 39.$$

42 Find the distance between the points $(-8, 5)$ and $(-14, 13)$.

- a. 9
- b. 10 ✓
- c. 14
- d. 11

$$\text{Distance} = \sqrt{(-14 - (-8))^2 + (13 - (5))^2} = \sqrt{(-6)^2 + (8)^2} = \sqrt{100} = 10.$$

43 Find the distance between the points $(1, -1)$ and $(25, -33)$.

- a. 39
- b. 41
- c. 40 ✓
- d. 56

$$\text{Distance} = \sqrt{(25 - (1))^2 + (-33 - (-1))^2} = \sqrt{(24)^2 + (-32)^2} = \sqrt{1600} = 40.$$

44 Find the distance between the points $(7, -4)$ and $(28, -76)$.

- a. 76
- b. 74
- c. 75 ✓
- d. 93

$$\text{Distance} = \sqrt{(28 - (7))^2 + (-76 - (-4))^2} = \sqrt{(21)^2 + (-72)^2} = \sqrt{5625} = 75.$$

45 Find the distance between the points $(-2, 6)$ and $(-11, -34)$.

a. 41 ✓

b. 40

c. 49

d. 42

$$\text{Distance} = \sqrt{(-11 - (-2))^2 + (-34 - (6))^2} = \sqrt{(-9)^2 + (-40)^2} = \sqrt{1681} = 41.$$

46 Find the distance between the points $(1, -1)$ and $(-8, 39)$.

a. 41 ✓

b. 42

c. 49

d. 40

$$\text{Distance} = \sqrt{(-8 - (1))^2 + (39 - (-1))^2} = \sqrt{(-9)^2 + (40)^2} = \sqrt{1681} = 41.$$

47 Find the distance between the points $(-6, 8)$ and $(-46, -34)$.

a. 58 ✓

b. 57

c. 82

d. 59

$$\text{Distance} = \sqrt{(-46 - (-6))^2 + (-34 - (8))^2} = \sqrt{(-40)^2 + (-42)^2} = \sqrt{3364} = 58.$$

48 Find the distance between the points $(3, -6)$ and $(15, -22)$.

a. 20 ✓

b. 19

c. 21

d. 28

$$\text{Distance} = \sqrt{(15 - 3)^2 + (-22 - (-6))^2} = \sqrt{(12)^2 + (-16)^2} = \sqrt{400} = 20.$$

49 Find the distance between the points $(-6, -4)$ and $(12, 20)$.

a. 31

b. 30 ✓

c. 29

d. 42

$$\text{Distance} = \sqrt{(12 - (-6))^2 + (20 - (-4))^2} = \sqrt{(18)^2 + (24)^2} = \sqrt{900} = 30.$$

50 Find the distance between the points $(7, -3)$ and $(10, -7)$.

a. 6

b. 7

c. 5 ✓

d. 4

$$\text{Distance} = \sqrt{(10 - 7)^2 + (-7 - (-3))^2} = \sqrt{(3)^2 + (-4)^2} = \sqrt{25} = 5.$$

51 Find the distance between the points $(-4, -3)$ and $(20, 42)$.

a. 51 ✓

b. 69

c. 50

d. 52

$$\text{Distance} = \sqrt{(20 - (-4))^2 + (42 - (-3))^2} = \sqrt{(24)^2 + (45)^2} = \sqrt{2601} = 51.$$

52 Find the distance between the points $(-7, 7)$ and $(9, 37)$.

a. 34 ✓

b. 33

c. 46

d. 35

$$\text{Distance} = \sqrt{(9 - (-7))^2 + (37 - (7))^2} = \sqrt{(16)^2 + (30)^2} = \sqrt{1156} = 34.$$

53 Find the distance between the points $(8, 0)$ and $(-12, -21)$.

a. 28

b. 41

c. 29 ✓

d. 30

$$\text{Distance} = \sqrt{(-12 - (8))^2 + (-21 - (0))^2} = \sqrt{(-20)^2 + (-21)^2} = \sqrt{841} = 29.$$

Multiple choice — hard

54 In a right triangle, the angle θ has an opposite side of length 7, adjacent side 24, and hypotenuse 25. Find $\cos \theta$.

a. $\frac{24}{7}$

b. $\frac{24}{25}$ ✓

c. $\frac{25}{24}$

d. $\frac{25}{7}$

$$\cos \theta = \text{adjacent/hypotenuse} = 24/25 = \frac{24}{25}.$$

- 55 In a right triangle, the angle θ has an opposite side of length 12, adjacent side 16, and hypotenuse 20. Find $\tan \theta$.

- a. $\frac{4}{3}$
- b. $\frac{5}{3}$
- c. $\frac{3}{4}$ ✓
- d. $\frac{5}{4}$

$$\tan \theta = \text{opposite/adjacent} = 12/16 = \frac{3}{4}.$$

- 56 In a right triangle, the angle θ has an opposite side of length 6, adjacent side 8, and hypotenuse 10. Find $\sin \theta$.

- a. $\frac{5}{3}$
- b. $\frac{3}{5}$ ✓
- c. $\frac{4}{3}$
- d. $\frac{5}{4}$

$$\sin \theta = \text{opposite/hypotenuse} = 6/10 = \frac{3}{5}.$$

- 57 In a right triangle, the angle θ has an opposite side of length 7, adjacent side 24, and hypotenuse 25. Find $\tan \theta$.

- a. $\frac{25}{7}$
- b. $\frac{25}{24}$
- c. $\frac{7}{24}$ ✓
- d. $\frac{24}{7}$

$$\tan \theta = \text{opposite/adjacent} = 7/24 = \frac{7}{24}.$$

- 58 In a right triangle, the angle θ has an opposite side of length 3, adjacent side 4, and hypotenuse 5. Find $\tan \theta$.

- a. $\frac{5}{3}$
- b. $\frac{4}{3}$
- c. $\frac{5}{4}$
- d. $\frac{3}{4}$ ✓

$$\tan \theta = \text{opposite/adjacent} = 3/4 = \frac{3}{4}.$$

- 59 A circle has radius 5. Find the sector area of a 180° sector, in terms of π .

- a. $\frac{23}{2}\pi$
- b. 25π
- c. $\frac{27}{2}\pi$
- d. $\frac{25}{2}\pi$ ✓

$$\text{Sector area} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{180}{360} \times \pi(25) = \frac{25}{2}\pi.$$

- 60 A circle has radius 11. Find the arc length of a 180° sector, in terms of π .

- a. 10π
- b. 22π
- c. 11π ✓
- d. 12π

$$\text{Arc length} = \frac{\theta}{360^\circ} \times 2\pi r = \frac{180}{360} \times 2\pi(11) = 11\pi.$$

61 A circle has radius 3. Find the arc length of a 45° sector, in terms of π .

- a. $\frac{1}{10}\pi$
- b. $\frac{3}{4}\pi$ ✓
- c. $\frac{3}{2}\pi$
- d. $\frac{7}{4}\pi$

$$\text{Arc length} = \frac{\theta}{360^\circ} \times 2\pi r = \frac{45}{360} \times 2\pi(3) = \frac{3}{4}\pi.$$

62 A circle has radius 5. Find the sector area of a 270° sector, in terms of π .

- a. $\frac{75}{2}\pi$
- b. $\frac{71}{4}\pi$
- c. $\frac{75}{4}\pi$ ✓
- d. $\frac{79}{4}\pi$

$$\text{Sector area} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{270}{360} \times \pi(25) = \frac{75}{4}\pi.$$

63 A circle has radius 7. Find the sector area of a 135° sector, in terms of π .

- a. $\frac{147}{8}\pi$ ✓
- b. $\frac{139}{8}\pi$
- c. $\frac{147}{4}\pi$
- d. $\frac{155}{8}\pi$

$$\text{Sector area} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{135}{360} \times \pi(49) = \frac{147}{8}\pi.$$

64 Triangle ABC is similar to triangle DEF , with $AB = 35$, $DE = 14$, and $BC = 35$. Find EF .

- a. 16
- b. 12
- c. 35
- d. 14 ✓

The scale factor is $DE/AB = 14/35 = \frac{2}{5}$. So $EF = \frac{2}{5} \times BC = \frac{2}{5} \times 35 = 14$.

65 Triangle ABC is similar to triangle DEF , with $AB = 12$, $DE = 18$, and $BC = 8$. Find EF .

- a. 14
- b. 8
- c. 10
- d. 12 ✓

The scale factor is $DE/AB = 18/12 = \frac{3}{2}$. So $EF = \frac{3}{2} \times BC = \frac{3}{2} \times 8 = 12$.

66 Triangle ABC is similar to triangle DEF , with $AB = 32$, $DE = 24$, and $BC = 12$. Find EF .

- a. 9 ✓
- b. 7
- c. 11
- d. 12

The scale factor is $DE/AB = 24/32 = \frac{3}{4}$. So $EF = \frac{3}{4} \times BC = \frac{3}{4} \times 12 = 9$.

67 Triangle ABC is similar to triangle DEF , with $AB = 21$, $DE = 7$, and $BC = 15$. Find EF .

- a. 5 ✓
- b. 15
- c. 7
- d. 3

The scale factor is $DE/AB = 7/21 = \frac{1}{3}$. So $EF = \frac{1}{3} \times BC = \frac{1}{3} \times 15 = 5$.

68 Triangle ABC is similar to triangle DEF , with $AB = 20$, $DE = 12$, and $BC = 15$. Find EF .

a. 15

b. 11

c. 7

d. 9 ✓

The scale factor is $DE/AB = 12/20 = \frac{3}{5}$. So $EF = \frac{3}{5} \times BC = \frac{3}{5} \times 15 = 9$.